

Planning a fair test

Change one thing. Trust the comparison.

Plan variables, record with units and judge what the evidence can support.

What will we change?

Does the run-out surface affect how far a trolley travels before stopping?

Name one thing to CHANGE. If ready, name two things to KEEP THE SAME.

Three steps in the investigation

- 1 Name the variables.
- 2 Plan three trials for each surface.
- 3 Judge what the results can tell us.

Watch an unfair comparison

The adult changes the surface AND adds a push.

The trolley stops farther away.

Which change caused that difference?

What would you change about the method?

Repair the method

CHANGE: run-out surface

KEEP: trolley, slope, release

MEASURE: stopping distance / cm

A variable is something that could change in the investigation.

Plan each box before testing.

Classify the five variables

The surface the trolley rolls on

Distance the trolley travels

Height of the ramp

Which trolley we use

How the trolley is released

Check the variable roles

CHANGE AND MEASURE

CHANGE: run-out surface
MEASURE: stopping distance / cm

KEEP THE SAME

Ramp height
Same trolley
Release mark; no push

Direct one useful correction

Check the ruler's zero against the agreed start.

Is the ruler end the same as its zero mark?

Direct the adult to correct one control.

Say the plan before touching kit

I will change _____. I will keep _____ and _____ the same. I will measure _____ in _____.

Your plan should make the comparison clear.

Fair method or a problem?

Same ramp height; different surfaces

Different ramp heights AND different surfaces

Three repeats for each surface

Ignoring a result you did not like

Measuring in 'about a ruler'

Check what each problem means

SUPPORTS THE COMPARISON

Same height; one planned change
Repeats with method unchanged

NEEDS CORRECTION

Two changed causes: unclear
Discarding disliked data: biased
'About a ruler': no clear unit

Keep an unexpected result

A trolley stopped much farther away on Trial 3. That is something to investigate.

Keep the value. Note any method issue. Plan a repeat.

Keep your plan for Tuesday

I will change the surface, control the release and measure stopping distance in cm.

Three repeats can show consistency. They cannot repair an unfair method.